Math 9 - Polynomials



— SINE TIMORE AUT FAVORE —

ST. GEORGE'S SCHOOL

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Last Updated: February 26, 2025

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1 Defining Polynomials

We will look at some vocabulary and terms associated with polynomials, as well as some modeling of polynomials with algebra tiles.

In mathematics, we commonly use a letter or symbol to represent a **variable** or unknown value.

Common Variables: $x, y, z, a, b, c, \theta, \alpha, \beta, \ldots$

The **coefficient** is the number that is being multiplied by a variable or variables.

Example 1.1 What are the variables and the coefficients of the following?

a) $-3x^2$ b) $15xy^2$ c) z^7

A **term** is either just a number or variables with coefficients.

Example 1.2 Determine the number of terms in each expression.

a) 5

b) $5x^2 + 5x + 5$

c) $5x + 5$

d) $7xyz - 6xy - 8z$

We refer to a term with no variable as a **constant**. The coefficient of a constant is the same value of the constant.

Example 1.3 Determine the constant value in each expression.

a) 5 b) $7x^2 + 3x - 10$ c) $4x + \pi - 3y$

A **monomial** is an expression of the form ax^n, where a is a **real number** and n is a **non-negative** integer. Monomials can have multiple variables as well.

Monomials	Non-Monomials
x	$\frac{1}{x}$
$\frac{4}{5}x^7$	$\frac{4}{5}x^{-7}$
xy^6	$\sqrt{x}y$
$\sqrt{6}$	$\sqrt{-1}x$

A **polynomial** is an expression of one or more monomials.

$$3x, 2x - 1, 4x^2 + y + 2, \ldots$$

Degrees and Leading Term

The **degree** of a term is the sum of the exponents of its variables.

The **leading term** of a polynomial is the term of highest degree. The degree of a polynomial is the same as the degree of its leading term.

Example 1.4 Identify the leading term and degree of the following polynomials.

a) $4x^2 + y + 2$

b) $6 + 4x^2 - 2xy^2$

Classifying Polynomials

We often refer to common polynomials by the number of terms it has.

Type	Examples
Monomial: A polynomial of one term	
Binomial: A polynomial of two terms	
Trinomial: A polynomial of three terms	

Beyond three terms, we usually just refer to them as a polynomial.

Notation

Polynomials are usually written in descending order of powers.

Example 1.5 Rewrite $2x + 6 - 7x^2$ in descending order of powers

If the degrees of the terms are equal, then we go with alphabetical order

Example 1.6 Rewrite $xy + 4y^2 - 7x^2$ in descending and alphabetical order

Algebra Tiles

We can use algebra tiles to model polynomials with one variable (usually x).



Example 1.7 Express the polynomial $x^2 + 2x - 3$ using algebra tiles.

Example 1.8 Express the polynomial $-2x^2 - x + 3$ using algebra tiles.

Simplifying Polynomials

Polynomial expressions can be simplified by combining like terms. **Like terms** are terms with the same variables raised to exactly the same powers.

Example 1.9 Simplify $4x^2 + 3x^2 - 5x^2$

Example 1.10 Simplify $7x + 5x^2 - 3x^2$

Example 1.11 Simplify $2xy^2 + 3x^2y - 4x^2y + xy^2$

Practice Exercises

1. Complete the table below for each polynomial.

Expression	Classification	Leading Coefficient	Degree
$2 - x^2$			
xy^2			
$6x - y^2$			
$x^2 + 3x + 2$			
$4xyz^2$			
8			
$x^2 - y^2 - 1$			

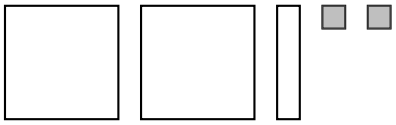
2. Write a polynomial with the following properties.

Classification	Leading Coefficient	Degree	Expression
Binomial	-4	6	
Trinomial	7	2	
Monomial	11	0	

3. Provide a reason why the following are **not** polynomials

Expression	Reason
$5x^2 + 3x + 4 - \frac{6}{x}$	
$\sqrt{6} + \sqrt{x}$	
$x^2 - x^{-2}$	
$\sqrt{-2}x^5$	

4. Write the polynomial expressed with the following algebra tiles. (shaded = positive)

a) 

b) 

5. Express the following using algebra tiles.

a) $2x^2 + 3x - 2$

b) $-x^2 - 2x + 5$

6. Simplify the following polynomials by collecting like terms.

a) $2c + 3 + 3c + 1$

b) $2x^2 + 3x - 5x$

c) $3f^2 + 3 - 6f^2 - 2$

d) $4b^2 - 2b + 5b + 4b^2 + 1$

e) $5t - 4 - 2t^2 + 3 + 6t^2$

f) $6 - 3x + x^2 + 9 - x$

g) $5m - 2m^2 - m^2 + 5m$

h) $4x^2 + 5y - 2x^2 - 1 - y$

i) $a^2 - 2a - 4 + 2a - a^2 + 4$

j) $5x - x^2 + 3x + x^2 - 7$

k) $pq - 1 - p^2 + 5p - 5pq - 2p$

l) $4gh + 7 - 2g^2 - 3gh - 11 + 6g$

m) $3r^2 - rs + 5s + r^2 - 2rs - 4s$

n) $-5s + st - 4s^2 - 12st + 10s - 2s^2$

2 Adding and Subtracting Polynomials

Adding and subtracting polynomials uses the same skills as adding and subtracting integers, except, we can only do this for like terms.

Using Algebra Tiles

Using algebra tiles to add is quite useful for visualization. Unfortunately, the same cannot be said with subtraction.

The basis of algebra tile addition is that like tiles combine, and opposite tiles cancel out.

Example 2.1 Add $(-2x^2 - 4) + (3x^2 - 2x + 2)$ using algebra tiles.

Adding without Algebra Tiles

This is the main method we want to focus on understanding. Two key skills when adding.

1. Identify like terms.
2. Add the integer coefficients and simplify to a single term

Example 2.2 Add and simplify $(-3x^2 + 4x - 5) + (6x^2 - 2x + 7)$

Vertical Addition

Sometimes we write our addition and subtraction statements vertically. This is commonly done when the like terms are already lined up vertically for us. We can then just focus on the integer addition.

Example 2.3 Add and simplify the following polynomials.

a)
$$\begin{array}{rr} & 2x^2+3x-8 \\ (+) & 5x^2+9x+7 \\ \hline \end{array}$$

b)
$$\begin{array}{rr} & -x^2-5x+8 \\ (+) & -6x^2+9x-9 \\ \hline \end{array}$$

Subtracting Polynomials

We'll spare the troubles of subtracting with algebra tiles. They work fine when you are only subtracting from what you have already. However, it is less intuitive when we need to subtract from things we don't have. Typically, this is done with net zero models.

So more often, we will usually just subtract polynomials horizontally or vertically.

Example 2.4 Subtract and simplify $(-3x^2+4x-5)-(6x^2-2x+7)$

Vertical Subtraction

Very important that we keep track of the signs and subtract properly.

Example 2.5 Subtract and simplify the following polynomials.

a)
$$\begin{array}{rr} & 2x^2+3x-8 \\ (-) & 5x^2+9x+7 \\ \hline \end{array}$$

b)
$$\begin{array}{rr} & -x^2-5x+8 \\ (-) & -6x^2+9x-9 \\ \hline \end{array}$$

Example 2.6 Add or subtract the following. Simplify where possible.

a) $(3 - 2x^2 + 4x) + (-2x + x^2 - 5)$

b) $(2xy + x - y) + (-yx - 2x - y)$

c) $(-4xy - x + 3y) + (-xy + 7y - 3)$

d) $(-5 + 3x^2 + 7x) + (-x^2 + 4 - 7x)$

e) $(-3x + 2x^2 + 4) - (-2 - x^2 + 4x)$

f) $(3x - 2x^2 + 4) - (-2x^2 + 3x - 4)$

g) $(4x - 5x^2 - 1) - (-3x^2 - 2x)$

h) $(2xy + 3y) - (y - x + xy)$

Practice Exercises

1. Add the following polynomials.

a) $(3x+2)+(-5x-3)$

b) $(5x-4)+(2x-10)$

c) $(-3x^2+6x)+(x^2-4x-3)$

d) $(x^2-3)+(x^2+3)$

e) $(5-2x^2+3x)+(-3x-4x^2+3)$

f) $(-3x+6-2x^2)+(-6+3x^2)$

g) $(3x-2x^2)+(-5+5x^2)$

h) $(2-3x^2+6x)+(-x^2-4x)$

i) $(4-2x^2)+(-5+x^2)$

j) $(-3xy-x^2+2y^2)+(-2xy-x^2-y^2)$

k) $(3x-2xy+2y)+(xy+1)+(2x-y)$

l) $(-2y+4x+xy)+(3xy-x)+(2y-x)$

2. Add the following polynomials.

a) $\begin{array}{r} -3x+1 \\ (+)\quad 2x-3 \\ \hline \end{array}$

b) $\begin{array}{r} -3x^2+4x \\ (+)\quad x^2+9x-2 \\ \hline \end{array}$

c) $\begin{array}{r} x^2-4 \\ (+)\quad x^2+4 \\ \hline \end{array}$

d) $\begin{array}{r} x^2+4x-5 \\ (+)\quad -2x^2-3x-4 \\ \hline \end{array}$

e) $\begin{array}{r} 2x^2+\ y^2-3xy \\ (+)\quad -3x^2+2y^2-4xy \\ \hline \end{array}$

f) $\begin{array}{r} 3x^2-2x+5 \\ (+)\quad -2x^2+6x-3 \\ \hline \end{array}$

g) $\begin{array}{r} 4x^2+3y-\ 8z \\ (+)\quad 2x^2-6y+10z \\ \hline \end{array}$

h) $\begin{array}{r} 4xy+\ x-y \\ (+)\quad 2x-xy+y \\ \hline \end{array}$

3. Subtract the following polynomials.

a) $(3x + 2) - (-5x - 3)$

b) $(5x - 4) - (2x - 10)$

c) $(-3x^2 + 6x) - (x^2 - 4x - 3)$

d) $(x^2 - 3) - (x^2 + 3)$

e) $(5 - 2x^2 + 3x) - (-3x - 4x^2 + 3)$

f) $(-3x + 6 - 2x^2) - (-6 + 3x^2)$

g) $(3x - 2x^2) - (-5 + 5x^2)$

h) $(2 - 3x^2 + 6x) - (-x^2 - 4x)$

i) $(4 - 2x^2) - (-5 + x^2)$

j) $(-3xy - x^2 + 2y^2) - (-2xy - x^2 - y^2)$

k) $(3x - 2xy + 2y) - (xy + 1) - (2x - y)$

l) $(-2y + 4x + xy) - (3xy - x) - (2y - x)$

4. Subtract the following polynomials.

a) $$\begin{array}{rr} & -3x+1 \\ (-) & 2x-3 \\ \hline \end{array}$$

b) $$\begin{array}{rr} & -3x^2+4x \\ (-) & x^2+9x-2 \\ \hline \end{array}$$

c) $$\begin{array}{rr} & x^2-4 \\ (-) & x^2+4 \\ \hline \end{array}$$

d) $$\begin{array}{rr} & x^2+4x-5 \\ (-) & -2x^2-3x-4 \\ \hline \end{array}$$

e) $$\begin{array}{rr} & 2x^2+y^2-3xy \\ (-) & -3x^2+2y^2-4xy \\ \hline \end{array}$$

f) $$\begin{array}{rr} & 3x^2-2x+5 \\ (-) & -2x^2+6x-3 \\ \hline \end{array}$$

g) $$\begin{array}{rr} & 4x^2+3y-8z \\ (-) & 2x^2-6y+10z \\ \hline \end{array}$$

h) $$\begin{array}{rr} & 4xy+x-y \\ (-) & 2x-xy+y \\ \hline \end{array}$$

3 Multiplying Polynomials

When multiplying polynomials, we combine two skills.

Exponent Law (Product of Powers)

$$x^m \cdot x^n = x^{m+n}$$

What is the simplified expression for $x^3 \cdot x^7$?

What is the product of $5 \times (-6)$?

Example 3.1 Multiply $(5x^3)(-6x^7)$

Example 3.2 Multiply $(-4x^3)(-2x)$

Example 3.3 Multiply $(-6)(x^9)$

Distributive Property

$$A(B + C) = A \cdot B + A \cdot C$$

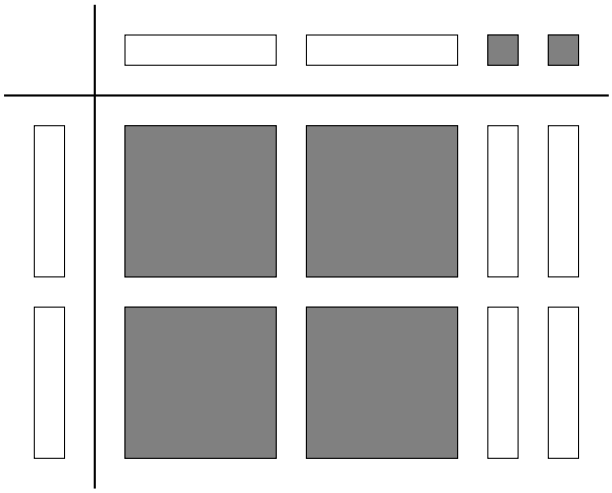
Example 3.4 Multiply $-4(3 - x)$

Example 3.5 Multiply $-3x(x^2 - 2x + 4)$

Multiplying with Algebra Tiles

We can model the product of two simple binomials or monomials with algebra tiles.

Example 3.6 Write the product statement from the model of the algebra tiles below. State the product as well. Positive tiles are shaded.



This is similar to area models for products.

Example 3.7 Represent 28×42 using an area model.

Example 3.8 Multiply the following.

a) $3(x^2 - 3x + 2)$

b) $-3x(x^2 - 2x - 4)$

c) $3x^2(2x^2 + 4x - 1)$

d) $-2xy(-x^2 + 2xy - y^2)$

e) $3xy^2(2x - 4y + 3z)$

f) $-2xyz(x^2 - y^2 + z^2)$

g) $2x^2y(-x + 3y - 2z)$

h) $-3y^2z(-2x + 3y - 2z)$

Practice Exercises

1. Find the product of the following.

   a) $3x \cdot 4x$

   b) $3x \cdot (-4x^2)$

   c) $-3x \cdot 4x^2$

   d) $-3x^2 \cdot (-4x^4)$

   e) $5xy \cdot 7xy$

   f) $5x^2 \cdot 7xy^2$

   g) $-5y \cdot (-7x)$

   h) $-5y^2 \cdot 7x^2y$

2. Find the product of the following.

   a) $3(x + 1)$

   b) $2(5 - x)$

   c) $-(x + 2)$

   d) $-(3 - x)$

   e) $4(a + 2b)$

   f) $3(2x + y)$

   g) $5(x - y)$

   h) $6(-x^2 + y^2)$

i) $-2(x+4)$

j) $-3(2x-1)$

k) $x(x+3)$

l) $2x(x-5)$

m) $-3(x+2)$

n) $-4(x-3)$

o) $-(7-x)$

p) $-2(x-y)$

q) $a(a+b)$

r) $-a(a-b)$

s) $x(2x-1)$

t) $2x(x^2-x-2)$

3. Find the product of the following.

a) $2xy(2x^2-3xy+7y^2)$

b) $-4pq(pq+3p^2+3q^2)$

c) $3g^2h(-2g+6h^2-3g^2-9g)$

d) $5s^2(-r^2+8rs-3s^2-4s+5r)$

4. For the below polynomials, express it as a product of a monomial and a polynomial. The first one is done for you.

a) $4x^3 + 8x^2 - 2x$
$= 2x(2x^2 + 4x - 1)$

b) $9x^2y^2 - 18xy$

c) $25x^2 - 10xy + 15xy^2$

d) $15x^2 - 9x - 12$

e) $16x^3y^3 - 20x^2y^2 + 36xy$

f) $-x^2y^2z^2 + 3xyz$

4 Dividing Polynomials

Division of polynomials can range in difficulty. We'll introduce division of polynomials by first dividing only by monomials.

Dividing monomials is the reverse of multiplying monomials, the coefficients divide, and the exponents subtract.

Example 4.1 Divide $12x^7 \div (-4x^3)$

Division Property

$$\frac{A+B}{C} = \frac{A}{C} + \frac{B}{C}$$

Can be extended to any number of terms in the numerator.

Example 4.2 Divide $\dfrac{6x^2 - 2x + 4}{2}$

Example 4.3 Divide $\dfrac{15x^2 - 9x - 12}{-3}$

Example 4.4 Divide $\dfrac{-8x^2 + 12xy}{4x}$

Example 4.5 Divide $\dfrac{-40x^2y + 10xy^2 - 5xy}{-5xy}$

Multiple Operations

We can simplify expressions with multiple operations using our standard order of operations (BEDMAS). Watch for "invisible" brackets, especially when subtracting a polynomial.

Example 4.6 Simplify the following.

a) $3x(x + 4) + \dfrac{9x^2 - 6x + 12}{3}$

b) $\dfrac{(6x - 9x^2)}{3x} - \dfrac{8x^2 - 12x}{4}$

Practice Exercises

1. Divide

a) $\dfrac{3x+6}{3}$

b) $\dfrac{3x-6}{3}$

c) $\dfrac{-3x+6}{3}$

d) $\dfrac{-3x-6}{3}$

e) $\dfrac{3x+6}{-3}$

f) $\dfrac{3x-6}{-3}$

g) $\dfrac{-3x+6}{-3}$

h) $\dfrac{-3x-6}{-3}$

i) $\dfrac{8x+12y}{4}$

j) $\dfrac{8x-12y}{4}$

k) $\dfrac{-8x+12y}{4}$

l) $\dfrac{-8x-12y}{4}$

m) $\dfrac{8x+12y}{-4}$

n) $\dfrac{8x-12y}{-4}$

o) $\dfrac{-8x+12y}{-4}$

p) $\dfrac{-8x-12y}{-4}$

2. Divide

a) $\dfrac{6x^2 - 12x + 18}{6}$

b) $\dfrac{6x^2 - 12x - 18}{6}$

c) $\dfrac{6x^2 - 12x + 18}{-6}$

d) $\dfrac{6x^2 + 12x - 18}{-6}$

e) $\dfrac{-6x^2 - 12x + 18}{6}$

f) $\dfrac{-6x^2 - 12x - 18}{6}$

g) $\dfrac{-6x^2 - 12x - 18}{-6}$

h) $\dfrac{-6x^2 - 12x + 18}{-6}$

i) $\dfrac{7x + 14y - 7}{7}$

j) $\dfrac{7x^2 + 14y^2 + 7}{7}$

k) $\dfrac{7x + 14y - 7}{-7}$

l) $\dfrac{7x^2 + 14y^2 + 7}{-7}$

m) $\dfrac{-7x + 14y + 7}{7}$

n) $\dfrac{-7x^2 + 14y^2 + 7}{-7}$

o) $\dfrac{36x^2 + 24xy - 28y^2}{-4}$

p) $\dfrac{25x^2 + 50xy - 10y^2}{-5}$

3. Divide

a) $\dfrac{5x^2+x}{x}$

b) $\dfrac{5x^2-x}{x}$

c) $\dfrac{-5x^2+2x}{x}$

d) $\dfrac{-5x^2-3x}{x}$

e) $\dfrac{-5x^2+2x}{-x}$

f) $\dfrac{-5x^2-3x}{-x}$

g) $\dfrac{12x^2+15xy}{3x}$

h) $\dfrac{12x^2-15xy}{3x}$

i) $\dfrac{12x^2+15xy}{-3x}$

j) $\dfrac{12x^2-15xy}{-3x}$

k) $\dfrac{12x^3+8x^2+4x}{4x}$

l) $\dfrac{-12x^3-8x^2+4x}{4x}$

m) $\dfrac{-12x^3+8x^2-4x}{-4x}$

n) $\dfrac{-12x^3-8x^2+4x}{-4x}$

o) $\dfrac{36x^3y^4+12x^2y^3-18x^2y^2}{6x^2y}$

p) $\dfrac{48x^4y^4-24x^3y^3-12x^2y^2}{-12x^2y^2}$

4. Perform the combined operations and then simplify.

a) $2x(x+3)+\dfrac{6x^2+4x-8}{2}$

b) $-\dfrac{(3x^2-6x+9)}{3}-4x(2-x)$

c) $\dfrac{-2x^2(3-x)}{2x}+\dfrac{6x(2x-4)}{2}$

d) $\dfrac{x(-6x^2+9x)}{3x}-\dfrac{6-4x}{2}$

e) $x(2x+3y-1)-\dfrac{4x^2+8xy-12x}{4}$

f) $\dfrac{-(4x^2-6x)}{2x}-3(1-x)$

g) $\dfrac{(8x^2+10x)}{2x}+\dfrac{4x(3-2x)}{2}$

h) $\dfrac{(12x-9x^2)}{3x}-\dfrac{(8x^2-16x)}{4}$

i) $\dfrac{-3x(8x-4x^2+6)}{-2}+\dfrac{6x(x-4)}{2}$

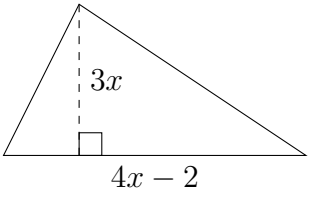
j) $\dfrac{-5x(3-6x)}{3}-\dfrac{4x(10x-5)}{5}$

5 Miscellaneous Skills

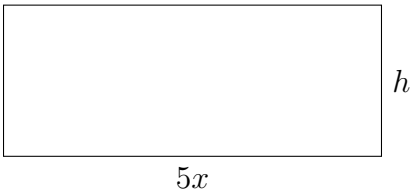
Confidently working with polynomials will quickly become one of our fundamental skills in mathematics.

The basic polynomial arithmetic that we are looking at now do not have natural application type questions, but the concepts of perimeter, area, and volume lend itself well to the operations we have practiced.

Example 5.1 Write a simplified expression for the area of the below triangle.



Example 5.2 Determine the expression for height, h, of the rectangle in terms of x, if the area is $10x^2 - 15x$.



Expanding Polynomials

Previously, we focused on products of one monomial by some other polynomial. However, we can and will often multiply polynomials with multiple terms by other polynomials with multiple terms. We will need to practice one key distributive concept.

$$(a+b)(c+d) = a(c+d) + b(c+d)$$

or

$$(a+b)(c+d) = ac + ad + bc + bd$$

Example 5.3 Expand and simplify $(x+3)(x+2)$.

Example 5.4 Expand and simplify $(2x+1)(3x-2)$.

Example 5.5 Expand and simplify.

a) $(x+3)(x-3)$

b) $(3x-5)(3x+5)$

Example 5.6 Expand and simplify.

a) $(3x+1)^2$

b) $(2x-3)^2$

Example 5.7 Expand and simplify $(x+2)(x-3)+5(x+4)$.

The expansion $(a+b)(c+d)$ can be referred to as a "2 by 2" expansion. It results in 4 terms, which can sometimes simplify.

Once you understand this process well, you can do other expansions such as "2 by 3" or "3 by 3".

Example 5.8 Expand and simplify.

a) $(2x+3)(x^2+4x+5)$

b) $(x^2-2x-1)(x^2+3x+2)$

If there are more than two things being multiplied at the same time, just choose two things to multiply first, and then multiply the remaining over it. The order does not matter because the order of multiplication does not matter.

Example 5.9 Expand and simplify.

a) $x(x+1)(x+2)$

b) $(x+1)(x-2)(x+2)$

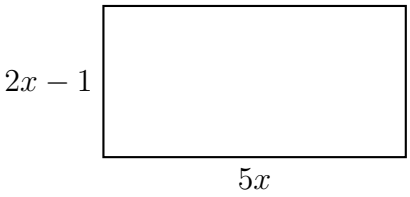
Practice Exercises

1. Write a simplified expression for the area for each the following rectangles.

a)



b)



2. Determine the expression for the height of a rectangle that has an area equal to $(24x^2 - 6x)$ and a base equal to $3x$.

3. Determine the expression for the base of a rectangle that has an area equal to $(8x^2 + 12x)$ and a height equal to $4x$.

4. Determine the expression for the height of a **triangle** that has an area equal to $(24x^2 - 6x)$ and a base equal to $3x$.

5. Determine the expression for the base of a **triangle** that has an area equal to $(8x^2 + 12x)$ and a height equal to $4x$.

6. Expand and simplify the following.

a) $(x+3)(x+7)$

b) $(x+5)(x-4)$

c) $(x-3)(x+6)$

d) $(x+2)(x-2)$

e) $(x-8)(x+3)$

f) $(2x+1)(3x+4)$

g) $(1-2x)(4x+1)$

h) $(4-x)(2x+3)$

i) $(3x-2)(1+2x)$

j) $(5-3x)(5+x)$

k) $(7-x)(4x+1)$

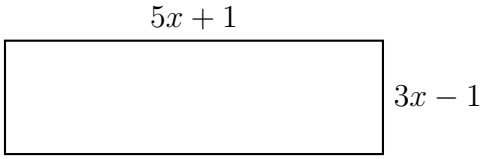
l) $(5x+2)^2$

m) $(3x-2)^2$

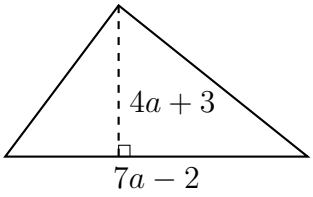
n) $(4x-5)^2$

7. Write a simplified expression for the area for each of the shapes below.

a)



b)



8. Expand and simplify the following.

a) $4(x-2)+(x+3)^2$

b) $-2(x-3)-(x+3)(x-3)$

c) $(1-3x)^2+(x+2)(x-3)$

d) $(1-x)^2+(x+2)^2$

e) $(x+3)(x^2+x+2)$

f) $(x+4)(x^2+x-2)$

g) $(x+2)(x^2+x+1)$

h) $(3x-2)(x^2-x-3)$

9. Expand and simplify the following.

a) $(x^2 + 2x + 3)(2x^2 - 3x - 1)$

b) $(2x^2 - x - 6)(4x^2 + 3x + 1)$

10. Expand and simplify the following.

a) $x(x - 4)(x + 1)$

b) $x(x - 3)(x - 2)$

c) $2x(x + 3)(x + 1)$

d) $-x(3 + x)(2 - x)$

e) $-3x(2x - 1)(x + 2)$

f) $2x^2(x - 1)^2$

g) $(x + 3)^3$

h) $(x - 1)^3$

i) $(2x + 1)^3$

j) $(3x - 2)^3$

6 Reasoning and Analyzing Problems

Each statement is either **always true, sometimes true,** or **never true**. When you decide on your answer, you must provide a **justification**.

If you choose **sometimes true**, you must show one scenario when it is true, and another, when it is false.

For **always true** and **never true**, you must provide a brief explanation why. Sometimes this can simply be a definition, or sometimes it may require some more mathematical reasoning.

Example 6.1 For each of the following statements, identify if the statement is always true, sometimes true, or never true. Provide a brief justification.

a) A polynomial that uses three variables is a trinomial.

b) If $3x^n + 1$ is a polynomial, then the degree of the polynomial is n.

c) If n is an integer, the expression $3x^n + x$ is a polynomial.

d) When multiplying a monomial by a simplified trinomial, the simplified result is a trinomial.

e) For all a, b, $(a + b)^2 = a^2 + b^2$.

f) If a is any number, then the polynomial expression, a^2, has degree 2.

Practice Exercises

1. For each of the following statements, identify if the statement is always true, sometimes true, or never true. Provide a brief justification.

   (a) If a simplified polynomial is a trinomial, then the degree is 2.

   (b) If the degree of a polynomial is 2, then it is a binomial.

   (c) If $ax^2 + bx^3$ is a polynomial, then the leading coefficient is b.

   (d) The product of two binomials is a trinomial.

   (e) When dividing a simplified binomial by a monomial, the simplified result is a monomial.

   (f) For all a, b, $(a + b)(a - b) = a^2 - b^2$.

   (g) If a polynomial is a constant, then the degree is 0.

7 Answer Key

Section 1 - Exercises

1. Complete the table below for each polynomial.

Expression	Classification	Leading Coefficient	Degree
$2 - x^2$	binomial	-1	2
xy^2	monomial	1	3
$6x - y^2$	binomial	-1	2
$x^2 + 3x + 2$	trinomial	1	2
$4xyz^2$	monomial	4	4
8	monomial or constant	8	0
$x^2 - y^2 - 1$	trinomial	1	2

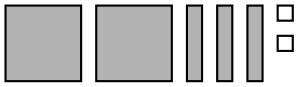
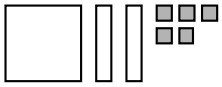
2. Expressions may vary.

Classification	Leading Coefficient	Degree	Expression
Binomial	-4	6	$-4x^6 + 2$
Trinomial	7	2	$7x^2 + 3x - 1$
Monomial	11	0	11

3. Reasons

Expression	Reason
$5x^2 + 3x + 4 - \frac{6}{x}$	$\frac{6}{x} = 6x^{-1}$, no negative exponents.
$\sqrt{6} + \sqrt{x}$	$\sqrt{x}$ is not allowed.
$x^2 - x^{-2}$	x^{-2} has a negative exponent.
$\sqrt{-2}x^5$	$\sqrt{-2}$ is not a real number.

4. a) $-2x^2 - x + 2$ b) $3x^2 - 3$

5. a)  b) 

6. a) $5c + 4$ b) $2x^2 - 2x$ c) $-3f^2 + 1$
   d) $8b^2 + 3b + 1$ e) $4t^2 + 5t - 1$ f) $x^2 - 4x + 15$
   g) $-3m^2 + 10m$ h) $2x^2 + 4y - 1$ i) 0
   j) $8x - 7$ k) $p^2 - 4pq + 3p - 1$ l) $-2g^2 + gh + 6g - 4$
   m) $4r^2 - 3rs + s$ n) $-6s^2 - 11st + 5s$

Section 2 - Exercises

1. a) $-2x - 1$ b) $7x - 14$ c) $-2x^2 + 2x - 3$
   d) $2x^2$ e) $-6x^2 + 8$ f) $x^2 - 3x$
   g) $4x^2 + 3x - 5$ h) $-4x^2 + 2x + 2$ i) $-x^2 - 1$
   j) $-2x^2 - 5xy + y^2$ k) $-xy + 5x + y + 1$ l) $4xy + 2x$

2. a) $-x - 2$ b) $-2x^2 + 13x - 2$ c) $2x^2$
   d) $-x^2 + x - 9$ e) $-x^2 + 3y^2 - 7xy$ f) $x^2 + 4x + 2$
   g) $6x^2 - 3y + 2z$ h) $3xy + 3x$

3. a) $8x + 5$ b) $3x + 6$ c) $-4x^2 + 10x + 3$
   d) -6 e) $2x^2 + 6x + 2$ f) $-5x^2 - 3x + 12$
   g) $-7x^2 + 3x + 5$ h) $-2x^2 + 10x + 2$ i) $-3x^2 + 9$
   j) $-xy + 3y^2$ k) $-3xy + x + 3y$ l) $-2xy + 6x - 4y$

4. a) $-5x + 4$ b) $-4x^2 - 5x + 2$ c) -8
   d) $3x^2 + 7x - 1$ e) $5x^2 - y^2 + xy$ f) $5x^2 - 8x + 8$
   g) $2x^2 + 9y - 18z$ h) $5xy - x - 2y$

Section 3 - Exercises

1. a) $12x^2$ b) $-12x^3$ c) $-12x^3$ d) $12x^6$
   e) $35x^2y^2$ f) $35x^3y^2$ g) $35xy$ h) $-35x^2y^3$

2. a) $3x+3$ b) $10-2x$ c) $-x-2$ d) $-3+x$
   e) $4a+8b$ f) $6x+3y$ g) $5x-5y$ h) $-6x^2+6y^2$
   i) $-2x-8$ j) $-6x+3$ k) x^2+3x l) $2x^2-10x$
   m) $-3x-6$ n) $-4x+12$ o) $-7+x$ p) $-2x+2y$
   q) a^2+ab r) $-a^2+ab$ s) $2x^2-x$ t) $2x^3-2x^2-4x$

3. a) $4x^3y-6x^2y^2+14xy^3$ b) $-4p^2q^2-12p^3q-12pq^3$
   c) $-6g^3h+18g^2h^3-9g^4h-27g^3h$ d) $-5s^2r^2+40rs^3-15s^4-20s^3+25rs^2$

4. a) $2x(2x^2+4x-1)$ b) $9xy(xy-2)$ c) $5x(5x-2y+3y^2)$
   d) $3(5x^2-3x-4)$ e) $4xy(4x^2y^2-5xy+9)$ f) $xyz(-xyz+3)$ or $-xyq(xyz-3)$

Section 4 - Exercises

1. a) $x+2$ b) $x-2$ c) $-x+2$ d) $-x-2$
   e) $-x-2$ f) $-x+2$ g) $x-2$ h) $x+2$
   i) $2x+3y$ j) $2x-3y$ k) $-2x+3y$ l) $-2x-3y$
   m) $-2x-3y$ n) $-2x+3y$ o) $2x-3y$ p) $2x+3y$

2. a) x^2-2x+3 b) x^2-2x-3 c) $-x^2+2x-3$
   d) $-x^2-2x+3$ e) $-x^2-2x+3$ f) $-x^2-2x-3$
   g) x^2+2x+3 h) x^2+2x-3 i) $x+2y-1$
   j) $x+2y^2+1$ k) $-x-2y+1$ l) x^2+2y^2+1
   m) $-x+2y+1$ n) x^2-2y^2-1 o) $-9x^2-6xy+7y^2$
   p) $-5x^2-10xy+2y^2$

3. a) $5x+1$ b) $5x-1$ c) $-5x+2$
   d) $-5x-3$ e) $5x-2$ f) $5x+3$
   g) $4x+5y$ h) $4x-5y$ i) $-4x-5y$
   j) $-4x+5y$ k) $3x^2+2x+1$ l) $-3x^2-2x+1$
   m) $3x^2-2x+1$ n) $3x^2+2x-1$ o) $6xy^3+2y^2-3y$
   p) $-4x^2y^2+2xy+1$

4. a) $5x^2+8x-4$ b) $3x^2-6x-3$ c) $7x^2-15x$
   d) $-2x^2+5x-3$ e) $x^2+xy+2x$ f) x
   g) $-4x^2+10x+5$ h) $-2x^2+x+4$ i) $-6x^3+15x^2-3x$
   j) $2x^2-x$

Section 5 - Exercises

1. a) $A = 5m^3 + 15m$ b) $A = 10x^2 - 5x$

2. $h = 8x - 2$ 3. $b = 2x + 3$ 4. $h = 16x - 4$ 5. $b = 4x + 6$

6. a) $x^2 + 10x + 21$ b) $x^2 + x - 20$ c) $x^2 + 3x - 18$
   d) $x^2 - 4$ e) $x^2 - 5x - 24$ f) $6x^2 + 11x + 4$
   g) $-8x^2 + 2x + 1$ h) $-2x^2 + 5x + 12$ i) $6x^2 - x - 2$
   j) $-3x^2 - 10x + 25$ k) $-4x^2 + 27x + 7$ l) $25x^2 + 20x + 4$
   m) $9x^2 - 12x + 4$ n) $16x^2 - 40x + 25$

7. a) $A = 15x^2 - 2x - 1$ b) $A = \dfrac{28a^2 + 13a - 6}{2} = 14a^2 + 6.5a - 3$

8. a) $x^2 + 10x + 1$ b) $-x^2 - 2x + 15$ c) $10x^2 - 7x - 5$
   d) $x^2 + 2x + 5$ e) $x^3 + 4x^2 + 5x + 6$ f) $x^3 + 5x^2 + 2x - 8$
   g) $x^3 + 3x^2 + 3x + 2$ h) $3x^2 - 5x^2 - 7x + 6$

9. a) $2x^4 + x^3 - x^2 - 11x - 3$ b) $8x^4 + 2x^3 - 25x^2 - 19x - 6$

10. a) $x^3 - 3x^2 - 4x$ b) $x^3 - 5x^2 + 6x$ c) $2x^3 + 8x^2 + 6x$
    d) $x^3 + x^2 - 6x$ e) $-6x^3 - 9x^2 + 6x$ f) $2x^4 - 4x^3 + 2x$
    g) $x^3 + 9x^2 + 27x + 27$ h) $x^3 - 3x^2 + 3x - 1$ i) $8x^3 + 12x^2 + 6x + 1$
    j) $27x^3 - 54x^2 + 36x - 8$

Section 6 - Exercises

1. a) Sometimes true. A trinomial can have a different degree than 2. ie. $x^3 + x^2 + x$ is a trinomial with degree 2. Whereas, $x^2 + x + 1$ is a trinomial with degree 2.

   b) Sometimes true. $x^2 + x + 1$ is a polynomial with degree 2, but it is a trinomial. $x^2 + 1$ is a binomial with degree 2.

   c) Always true. Because the degree is 3, then the leading coefficient of x^3 is b.

   d) Sometime true. It depends on if the middle terms simplify. ie. $(x + 3)(x + 3) = x^2 + 6x + 9$ which has three terms, whereas, $(x + y)(x + 3) = x^2 + 3x + xy + 3y$ has four terms.

   e) Never true. When dividing a simplified binomial by a monomial, the result must be a binomial. You should be able to multiply the result with the monomial to get back to the original binomial.

   f) Always true. The middle terms cancel. This is a difference of squares.

   g) Always true. A constant term has a degree of 0.